Gürsel TÜRKERİ

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Education

GAZIANTEP UNIVERSITY Gaziantep, TURKIYE
Master of Science in Mechanical Engineering 02/2026 - Present
Machine Theory and Dynamics

- Robotics, Mechatronic Systems, Optimization, Mechanism Kinematics

Bachelor of Mechanical Engineering (%100 English) 09/2019 – 06/2024

- GPA: 2.92/4.00

Experience

Kuartis Ankara, TURKIYE
Autonomous Driving Field Application Engineer 09/2024 – Present
CVMLR (Computer Vision, Machine Learning, Robotics)

- Conducting development, simulation, and field tests for autonomous land and sea vehicles using ROS2 and C++ on Linux-based platforms.
- Performed intrinsic camera calibration and multi-sensor data collection (cameras, LiDAR, GNSS) for perception and localization pipelines.
- Developed a mission management UI in Qt (C++) for autonomous vehicle operation, monitoring, and telemetry visualization.
- Managed containerized deployment and version control workflows using Docker and Git for reproducible ROS2 builds across embedded targets.

Baykar Technologies Istanbul, TURKIYE
Long Term Intern
02/2024 – 05/2024

- Built an autonomous fixed-wing flight simulation in Gazebo integrated with the PX4 autopilot via MAVROS and ROS2 to analyze guidance and control algorithms.
- Improved fixed-wing guidance algorithms to achieve more accurate flight-path tracking in autonomous PX4-based flight tests.
- Derived and implemented a mathematical model of the flare (landing) phase in MATLAB and validated it in simulation.

Middle East Technical University, ROMER Ankara, TURKIYE
Summer Intern
07/2023 – 08/2023

- Contributed to the ChessMate project using the Franka Emika Panda 7-DOF manipulator, integrated in ROS1 and simulated in Gazebo.
- Implemented motion planning and inverse-kinematics solving with MoveIt for precise pick-and-place manipulation.
- Improved a YOLOv8-based vision model for chess-piece recognition to enhance object detection and manipulation accuracy.

Automobile Robot and Energy Community (ORET) Gaziantep, TURKIYE
Team Member
10/2021 - 05/2023

- Developed an autonomous driving stack on the Autoware Universe framework (ROS2, C++, Python), covering perception, localization, planning, and control.
- Implemented PCD-based localization (NDT scan matching) and behavior path planning within the Autoware pipeline.
- Trained and deployed a YOLOv8 traffic-sign detection model, optimized with TensorRT on NVIDIA Jetson Xavier for real-time inference.
- Integrated stereo-camera detection with depth estimation using OpenCV for 3D object localization.

Skills & Interests

Technical: Python, C++, ROS2/ROS, Autoware, MoveIt, PX4/MAVROS, Gazebo, Docker, Git, Linux, YOLOv8, TensorRT, OpenCV, NVIDIA Jetson, Qt, Sensor Calibration & Fusion, MATLAB, Machine Learning, SOLIDWORKS
Language: Native in Turkish and working proficiency in English.